1. to apply Simpson's 1/3 rule, the number of intervals in the following must be

Answer: (select your correct answer)

6

7

9

11

2. In integrating by dividing the interval into eight equal parts, width of the interval should be

Answer: (select your correct answer)

0.125

0.250

0.500

0.625

3. ------lies in the category of iterative method.

Answer: (select your correct answer)

None of the given option

Bracketing method

Regula falsi method

Muller's method

4. Regula falsi method lies in the category of -------

Answer: (select your correct answer)

Iterative method

Bracketing method

Random method

Graphical method

5. It can be verified that for matrix

Answer: (select your correct answer)

AA^{-1}=I, I= identity matrix

AA^{-1}=D, D= diagonal matrix

AA^{-1}=S, S= symmetric matrix

AA^{-1}=Z, Z= orthogonal matrix

6. Newton's divided difference interpolation formula is used when the values of the independent variable are

Answer: (select your correct answer)

Equally spaced

Not equally spaced

Constant

None of the above

7. If $f(x) = 5x^6 + 6x^5 - 7x^3 - 9x^2 + 4x - 3$, then it's ----- derivative is zero for all x.

Answer: (select your correct answer)

4th

7th

6th

5th

8. to apply Simpson's 1/3 rule, valid number of intervals are ----

Answer: (select your correct answer)

7

8

5

3

9. In integrating by dividing into eight equal parts width of the interval should be

Answer: (select your correct answer)

0.250

0.500

0.125

0.625

H=b-1/n….trapezoidal

10. A fourth order ordinary differential equation can be reduced to a system of four ----- order ordinary differential equations.

Answer: (select your correct answer)

First

Second

Third

Fourth

11. In Richardson`s extrapolation method, the extrapolation process is repeated until accuracy is achieved, this is called extrapolation to the -----

Answer: (select your correct answer)

Limit

Function

Arbitrary value of "h"

None of given option

12. From the following table of values:

X	1.00	1.05	1.10	1.15	1.20	1.25	1.30
Y	1.0000	1.0247	1.0488	1.0724	1.0954	1.1180	1.1402

Answer: (select your correct answer)

Forward difference operator

Backward difference operator

Enter difference operator

None of the given option

7. While deriving Simpson`s 3/8 rule, we approximate f(x) in the form -----

Answer: (select your correct answer)

Ax+b

$Ax^2 + bx + c$

$Ax^3 + bx^2 + cx + d$

$Ax^4 + bx^3 + cx^2 + dx + e$

13. When we apply Simpson`s 3/8 rule, the number of intervals n must be

Answer: (select your correct answer)

Even

Odd

Multiple of 3

Multiple of 8

14. Next approximation to the root of the given equation by bisection method can be found if

Answer: (select your correct answer)

F(2) =-7, f(3)= -8

F(2) =-7, f(3)= 8

F(2) =7, f(3)= 8

None of the given option

15. if , then

Answer: (select your correct answer)

$S^{-1} = S^T$

$S^T = S$

I = S

None of given option

16. Lagrange`s interpolation formula is used when the values of the independent variable are

Answer: (select your correct answer)

Equally spaced

Not equally spaced

Constant

None of the above

17. If there are (n+1) values of y corresponding to (n+1) values of x, then we can represent the function f(x) by a polynomial of degree

Answer: (select your correct answer)

n+2

n+1

n

n-1

18. While deriving trapezoidal rule, we approximate f(x) in the form

Answer: (select your correct answer)

Ax+b

$Ax^2 + bx + c$

$Ax^3 + bx^2 + cx + d$

$Ax^4 + bx^3 + cx^2 + dx + e$

19. Two segment trapezoidal rule of integration is exact for integrating at most ---- order polynomial.

Answer: (select your correct answer)

First

Second

Third

Fourth

20. A third order ordinary differential equation can be reduced to a system of ---- first order ordinary differential equations.

Answer: (select your correct answer)

0

1

2

3

The first lngrange polynomial with equally spaced nodes produced the formula for

_____.

Simpson's rule

Trapezoidal rule

Newton's method

Richardson's method

While employing Trapezoidal and Simpson Rules to evaluate the double integral numerically, by using Trapezoidal and Simpson rule with respect to -------

- variable/variables at time **single**

The Trapezoidal Rule is an improvement over using rectangles because we have much less "missing" from our calculations.

We used _____ to model the curve in trapezoidal Rule

straight lines

curves

parabolas
constant

Adams - Bashforth is a multistep method.

True
False

To evaluate a definite integral of tabular function
f(x), piecewise quadratic approximation led to -----
—
Simpson's Rule

Which of the following is the Cote‟s number (weighting coefficient) for the function: f(x) = x+1 in the interval [0,1]?
1/2

In Simpson‟s rule, we can estimate the integral by
…… the areas under the parabolic arcs through three successive points.
Adding

In Newton-Cotes formula for finding the definite integral of a tabular function, which of the following is taken as an approximate function then find the desired integral?
Trigonometric Function
Exponential Function Logarithmic Function
Polynomial Function Trapezoidal and Simpson's integrations are just a linear combination of values of the given function at different values of the ……variable.
Dependent
Independent
Arbitrary
None of the given choices

The percentage error in numerical integration is defined as
= (Theoretical Value-Experiment Value)* Experiment Value*100

Simpson's 3/8 rule is based on fitting …… points by a cubic.
Two
Three
Four
None of the given choices

We can improve the accuracy of trapezoidal and
Simpson's rules using ……
Simpson's 1/3 rule Simpson's 3/8 rule
Richardson's extrapolation method
None of the given choices

In the process of Numerical Differentiation, we differentiate an interpolating polynomial in place of -----
——.
Newton's Divided Difference Interpolating polynomial

Exact solution of 2/3 is not exists.

TRUE

FALSE

A 3 x 3 identity matrix have three and _____Eigen values.

Same

Different

Eigen values of a symmetric matrix are all _____ .

Real

Complex

Zero

Positive

The Jacobi iteration converges, if A is strictly diagonally dominant.

TRUE

FALSE

Below are all the finite difference methods EXCEPT _____.

Jacobi's method

Newton's backward difference method

Stirlling formula

Forward difference method

If n x n matrices A and B are similar, then they have the same eigenvalues (with the same multiplicities).

TRUE

FALSE

If A is a nxn triangular matrix (upper triangular, lower triangular) or diagonal matrix, the eigenvalues of A are the diagonal entries of A.
TRUE

FALSE

The characteristics polynomial of a 3x 3

Identity matrix is _____, if x is the Eigen values of the given 3 x 3 identity matrix. Where symbol ^ shows power.

(X-1)^3

(x+1)^3

X^3-1

X^3+1

Two matrices with the same characteristic polynomial need not be similar.

TRUE

FALSE

The Jacobi's method is a method of solving a matrix equation on a matrix that has ____ zeros along its main diagonal.

no

atleast one

Bisection method is a

Bracketing method

Open method

Regula Falsi means

Method of Correct position

Method of unknown position

Method of false position

Method of known position

An eigenvector V is said to be normalized if the coordinate of largest magnitude is equal to zero.

Select correct option:

TRUE

FALSE

The Gauss-Seidel method is applicable to strictly diagonally dominant or symmetric
_____ definite matrices A.

Select correct option:

Positive

Negative

Differences methods find the _____ solution of the system.

Select correct option:

Numerical

Analytical

The Power method can be used only to find the eigenvalue of A that is largest in absolute value—we call this Eigenvalue the dominant eigenvalue of A.

Select correct option:

TRUE

FALSE

The Jacobi's method is a method of solving a matrix equation on a matrix that has no zeros along its _____.

Select correct option:

Main diagonal

Last column

Last row

First row

A 3 x 3 identity matrix have three and different Eigen values.

Select correct option:

TRUE

FALSE

Newton Raphson method falls in the category of

Bracketing method

Open Method

Iterative Method

Indirect Method

Newton Raphson method is also known as

Tangent Method

Root method

Open Method

Iterative Method

Secant Method uses values for approximation

1

3

2

4

Secant Method is than bisection method for finding root

Slow

Faster

In Newton Raphson method

Root is bracketed

Root is not bracketed

Regula-falsi method and bisection method are both

Convergent

In bisection method the two points between which the root lies are

Similar to each other
Different

Not defined

Opposite

In which methods we do not need initial approximation to start

Indirect Method

Open Method

Direct Method

Iterative Method

Root may be

Complex

Real

Complex or real

None

In Regula-falsi method we choose points that have signs

2 points opposite signs

3 points opposite signs

2 points similar signs

None of the given

In a bounded function values lie between

1 and -1

1 and 2

0 and 1

0 and -2

Newton Raphson method is a method which when it leads to division of number close to zero

Diverges

Converges

Which of the following method----- is modified form of Newton Raphson Method?

Regula-falsi method

Bisection method

Secant method

Jacobi's Method

Which 1 of the following is generalization of Secant method?

Muller's Method

Jacobi's Method

Bisection Method

N-R Method

Secant Method needs starting points

2

3

4

1

Near a simple root Muller's Method converges than the secant method

Faster

Slower

If we retain r+1 terms in Newton's forward difference formula, we

obtain a polynomial of degree ---- agreeing with y_x **at** $x_0, x_1, ..., x_r$

r+2

r+1

R

R-1

Octal numbers has the base

10

2

8

16

Newton's divided difference interpolation formula is used when the values of the independent variable are

Equally spaced

Not equally spaced

Constant

None of the above

Given the following data

x	0	1	2	4
$f(x)$	1	1	2	5

Value of $f(2.4)$ is

3

If $y(x)$ is approximated by a polynomial $p_n(x)$ of degree n then the error is given by

$\varepsilon(x)=y(x)-P_n(x)$

Let I the closed interval spanned by . Then vanishes ------times in the interval I denotes

N-1

N+2

N

N+1

To apply Simpson's 1/3 rule, valid number of intervals are.....

8 (bc it is even)

Finding the first derivative of $f(x)$ from the following table: at x =0.4

x	0.1	0.2	0.3	0.4
$f(x)$	1.10517	1.22140	1.34986	1.49182

Differential operator in terms of ----------will be used.

Forward difference operator

Backward difference operator

Central difference operator

All of the given choices

To apply Simpson's 1/3 rule, the number of intervals in the following must be

2 (**Simpson**"s **1/3 rule must use** an even **number** of elements')

3

5

7

To apply Simpson's 3/8 rule, the number of intervals in the following must be

10

11

12 (bc it is divisible by 3)

13

If the root of the given equation lies between a and b, then the first approximation to the root of the equation by bisection method is ……

$\frac{(a+b)}{2}$

$\frac{(a+b)}{2}$

$\frac{(b+a)}{2}$

None of the given choices

For the equation $x^3+3x+1=0$ the root of the equation lies in the interval……

(0, 1)

For the given table of values

x	0.1	0.2
$f(x)$	0.425	0.475

0.3	**0.4**	**0.5**	**0.6**
0.400	**0.452**	**0.525**	**0.575**

$f'(0.1)$ **, using two-point equation will be calculated as……**

-0.5

0.5

0.75

-0.75

For the given table of values

x	**0.1**	**0.2**	**0.3**	**0.4**	**0.5**	**0.6**
$f(x)$	**0.425**	**0.475**	**0.400**	**0.452**	**0.525**	**0.575**

$f''(0.2)$ **, using three-point equation will be calculated as ……**

17.5

12.5

7.5

-12.5

Rate of change of any quantity with respect to another can be modeled by

An ordinary differential equation

A partial differential equation

A polynomial equation

None of the given choices

If

$$\frac{dy}{dx} = f(x, y)$$

Then the integral of this equation is a curve in

None of the given choices

Xt-plane

Yt-plane

Xy-plane

In solving the differential equation

$$y' = x + y\,; y(0.1) = 1.1$$

$h = 0.1$ **, By Euler's method** $y(0.2)$ **is calculated as**

1.44

1.11

1.22

1.33

Adam-Moulton P-C method is derived by employing

Newton's backward difference interpolation formula

Newton's forward difference interpolation formula

Newton's divided difference interpolation formula

None of the given choices

The need of numerical integration arises for evaluating the definite integral of a function that has no explicit _____ or whose antiderivative is not easy to obtain

Derivatives

Antiderivative

If $A \neq 0$ then system will have a

Definite solution

Unique solution

Correct solution

No solution

If $A \neq 0$ then

There is a unique solution

There exists a complete solution

There exists no solution

None of the above options

Direct method consists of method

2 (elimination and decomposition)

3

5

4

We consider Jacobi's method Gauss Seidel Method and relaxation method as

Direct method

Iterative method

Open method

All of the above

In Gauss Elimination method Solution of equation is obtained in

3 stages

2 stages

4 stages

5 stages

Gauss Elimination method fails if any one of the pivot values becomes
Greater

Small

Zero

None of the given

Changing the order of the equation is known as

Pivoting

Interpretation

Full pivoting is than partial pivoting

Easy

More complicated

The following is the variation of Gauss Elimination method

Jacobi's method

Gauss Jordan Elimination method

Courts reduction method is also known as Cholesky Reduction method

True

False

Jacobi's method is also known as method of Simultaneous displacement

True

False

Bisection and false position methods are also known as bracketing method and are always

Divergent **=>Convergent**

The Inverse of a matrix can only be found if the matrix is

Singular Scalar Diagonal **=>Non-singular**

If f (x) contains trigonometric, exponential or logarithmic functions then this equation is known as

=>Transcendental Equation Algebraic Polynomial Linear

In interpolation is used to represent the δ Forward difference?

=>Central difference Backward difference

The base of the decimal system is ----

=>10 0 2 8 None of the above.

The determinant of a diagonal matrix is the product of the diagonal elements. **=>True**

False

Power method is applicable if the eigen vectors corresponding to eigen values are linearly independent.

=>True false

If n x n matrices A and B are similar, then they have the different eigenvalues (with the same multiplicities).

True **=>False**

An eigenvector V is said to be normalized if the coordinate of largest magnitude is equal to -

=>Unity zero

The determinant of a ---- matrix is the product of the diagonal elements.

=>Diagonal Upper triangular Lower triangular Scalar

The Power method can be used only to find the Eigen value of A that is largest in absolute value—we call this Eigen value the dominant Eigen value of A. **=>True** False

For differences methods we require the set of values. **=>True** False

If x is an Eigen value corresponding to Eigen value of V of a matrix A. If a is any constant, then x - a is an Eigen value corresponding to Eigen vector V is an of the matrix A - a I. **=>True** False

Central difference method seems to be giving a better approximation, however it requires more computations.

True **=>False**

Iterative algorithms can be more rapid than direct methods. **=>True** False

Central Difference method is the finite difference method. **=>True** False

If A is a n x n triangular matrix (upper triangular, lower triangular) or diagonal matrix , the eigenvalues of A are the diagonal entries of A. **=> TRUE** FALSE

Q: While solving by Gauss-Seidel method, which of the following is the first Iterative solution for the system;
x-2y =1,
x+4y=4 ?

=>(1,0.75) (0,0) (1,0) (0,1)

Q: While solving a system of linear equations by Gauss Jordon Method, after all the elementary row operations if there lefts also zeros on the main diagonal then which of the is true about the system?

System may have unique solutions

System has no solution

System may have multiple numbers of finite solutions

System may have infinite many solutions

(because **Gauss-Jordan Reduction of a matrix for a system can reveal if the system has :**

- **Unique Solution : Look for all *non-zero* values down the main *diagonal*.**
- **Infinite Solution : Bottom row(s) are *all zeroes*.**
- **No Solution : A bottom row implies a contradiction, i.e. *<zeroes> = <nonzero>***

Q: Which of the following method is not an iterative method?

Jacobi's method Gauss-Seidel method Relaxation methods **=>Gauss-Jordan elimination method**

Q: Numerical methods for finding the solution of the system of equations are classified as direct and …… methods:::::::; Indirect **=>Iterative** Jacobi None of the given choices

Q: If the Relaxation method is applied on the system; 2x+3y = 1, 3x +2y = - 4, then largest residual in 1st iteration will reduce to -------.

Zero **=>4** -1 -1

In …… method, the elements above and below the diagonal are simultaneously made zero.

Select correct option:

Jacobis Gauss-Seidel **=>Gauss-Jordon Elimination** Relaxation

If the order of coefficient matrix corresponding to system of linear equations is 3*3 then which of the following will be the orders of its decomposed matrices; 'L' and 'U'?

Select correct option:

=>Order of 'L' = 3*1, Order of 'U' = 1*3 Order of 'L' = 3*2, Order of 'U' = 2*3

Order of 'L' = 3*3, Order of 'U' = 3*3 Order of 'L' = 3*4, Order of 'U' = 4*3

Which of the following is equivalent form of the system of equations in matrix form;
AX=B?

Select correct option:

XA=B **=>X=B(Inverse of A)** X =(Inverse of A)B BX=A

Which of the following rearrangement make strictly diagonal dominant, the system of linear equations; x-3y+z= -2, -
6x+4y+11z=1, 5x-2y-2z=9?

5x-2y-2z=9, x-3y+z= -2, -6x+4y+11z=1 =>-6x+4y+11z=1, x-3y+z= -2, 5x-2y-2z=9

=>5x-2y-2z=9, -6x+4y+11z=1, x-3y+z= -2

No need to rearrange as system is already in diagonal dominant form.

If the determinant of a matrix A is not equal to zero then the system of equations will have…….
=>a unique solution many solutions infinite many solutions None of the given choices

Sparse matrix is a matrix with …….

Some elements are zero **=>Many elements are zero**

Some elements are one Many elements are one

Which of the following is the meaning of partial pivoting while employing the row transformations?

Select correct option:

=>Making the largest element as pivot Making the smallest element as pivot

Making any element as pivot Making zero elements as pivot

While solving the system; x-2y = 1, x+4y = 4 by Gauss-Seidel method, which of the following ordering is feasible to have
good approximate solution?:

x+4y = 1, x-2y = 4 x+2y = 1, x- 4y =4 x+4y = 4, x-2y = 1 **=>no need to reordering**

Back substitution procedure is used in

……

=>Gaussian Elimination Method Jacobi's method

Gauss-Seidel method None of the given choices

The linear equation: 2x+0y-2=0 has -------- solution/solutions.

=>unique **no solution** infinite many finite many

If a system of equations has a property that each of the equation possesses one large coefficient and the larger coefficients in the equations correspond to different unknowns in different equations, then which of the following iterative method id preferred to apply?

Gauss-Seidel method Gauss-Jordon method **=>Gauss elimination method** Crout's method

When the condition of diagonal dominance becomes true in Jacobi's Method.Then its means that the method is

Select correct option:

Stable Unstable **=>Convergent** Divergent

For a system of linear equations, the corresponding coefficient matrix has the value of determinant; |A| = 0, then which of the following is true?

The system has unique solution The system has finite multiple solutions

The system has infinite may solutions =>The system has no solution

For the system; 2x+3y = 1, 3x +2y = - 4, if the iterative solution is (0,0) and 'dxi = 2' is the increment in 'y' then which of the following will be taken as next iterative solution?

(2.0) (0.3) (0.2) (1.-4)

If system of equations is inconsistent then its means that it has

……

=>No Solutions Many solutions Infinite Many solutions None of the given choices

While using Relaxation method, which of the following is the Residuals for 1st iteration on the system; 2x+3y = 1, 3x +2y
=4

(2,3) (3,-2) (-2,3) **=>(1,4)**

Relaxation Method is a/an ……. Direct method **=>Iterative method**

Gauss - Jordan Method is similar to …….

Gauss-Seidel method Iteration's method Relaxation Method **=>Gaussian elimination method**

Full pivoting, in fact, is more ……..……than the partial
pivoting.

Easiest **=>Complicated**

Gauss-Seidel method is also known as method of
……..……

Successive displacement => Iterations False position None of the given choices

The Jacobi iteration ---, if A is strictly diagonally dominant.

=>converges diverges

Power method is applicable if the eigen vectors corresponding to eigen values are
linearly ----.:

=>independent dependent

Power method is applicable if the eigen values are -------.

=>real and distinct real and equal positive and distinct negative and distinct

How many Eigen vectors will exist corresponding to the function; Exp(ax) = e^ax, when the matrix
operator is of
differentiation?

=>Infinite many Unique Finite Multiple None

By using determinants, we can easily check that the solution of the given system of linear equation ---
and it is --

=>exits, unique exists, consistent trivial, unique nontrivial, inconsistent

Eigenvectors of a symmetric matrix are orthogonal, but only for distinct
eigenvalue.

⇒TRUE FALSE

Two matrices with the ---- characteristic polynomial need not be
similar.

⇒same different

The absolute value of a determinant (|detA|) is the product of the absolute values of the eigenvalues of matrix A

⇒TRUE **FALSE**

The eigenvectors of a square matrix are the non-zero vectors that, after being multiplied by the matrix, remain ……
to the original vector.

Perpendicular Parallel **⇒Diagonal** None of the given choices

In Jacobi's method after finding D1, the sum of the diagonal elements of D1 should be …… to the sum of the diagonal
elements of the original matrix A.

Greater than Less than **⇒Same** Different

Gauss Seidel method is also known as method of Successive displacement

False

True

In Jacobi's method approximation calculated is used for
Nothing

Calculating the next approximation

Replaced by previous one

All above

In Gauss Seidel method approximation calculated is replaced by previous one

True

False

Relaxation method is derived by

South well

Not defined

Power method is applicable for only

Real metrics

Symmetric

Unsymmetrical

Both symmetric and real

The process of eliminating value of y for intermediate value of x is know as interpolation

True

False

Computer uses the words that are

Infinite

Finite

Differences methods are iterative methods.
TRUE
FALSE

Eigen values of a _____ matrix are all real.

symmetric
antisymmetric
rectangular
triangular

Question : While solving a system of linear equations, which of the following approach is economical for the computer memory?
Select correct option:

Direct
Iterative
Analytical
Graphical

Question : The basic idea of relaxation method is to reduce the largest residual to ……

One
Two
Zero
None of the given choices

Question : Which of the following is a reason due to which the LU decomposition of the system of linear

equations; x+y
= 1, x+y =2 is not possible?

Associated coefficient matrix is singular
All values of l's and u's can't be evaluated
Determinant of coefficient matrix is zero
All are equivalent

Question : While using Relaxation method, which of the following is the largest Residual for 1st iteration on the system;
2x+3y = 1, 3x +2y = - 4 ?

-4
3
2
1

Question : Jacobi's Method is a/an……
Iterative method
Direct metho

Question: In …… method, a system is reduced to an equivalent diagonal form using elementary transformations.

Jacobi's
Gauss-Seidel
Relaxation
Gaussian elimination

Question : Under elimination methods, we consider, Gaussian elimination and ……methods. :

Gauss-Seidel
Jacobi
Gauss-Jordan elimination
None of the given choices

Page No.72

Question : Gauss-Seidel method is similar to …….
Iteration's method
Regula-Falsi method
Jacobi's method
None of the given choices

Page No.67

.The Gauss-Seidel method is applicable to strictly diagonally dominant
or symmetric positive definite matrices A.
True
False

Question : Simpson's rule is a numerical method that approximates the value of a definite integral by
using polynomials.
Quadratic
Linear
Cubic
Quartic

Question : .In Simpson's Rule, we use parabolas to approximating

each
part of the curve. This proves to be very efficient as compared to Trapezoidal rule.

True

False

Question : The predictor-corrector method an implicit method. (multi-step methods)

True

False

Question : Generally, Adams methods are superior if output at many points is needed.

True

False

Question : The Trapezoidal rule is a numerical method that approximates the value of a._____.

Indefinite integral

Definite integral

Improper integral

Function

Question : .An indefinite integral may _____ in the sense that the limit defining it may not exist.

diverge

Converge

Questi on : An improper integral is the limit of a definite integral as an endpoint of the interval of integratio n approaches either a specified real number or ∞ or -∞ or, in some cases, as both endpoints approach limits.

TRUE

FALSE

Question : Euler's Method numerically computes the approximate derivative of a function.

TRUE

FALSE

Question :.Euler's Method numerically computes the approximate _____ of a function.

Antiderivative

Derivative

Error

Value

If we wanted to find the value of a definite integral with an infinite limit, we can instead replace the infinite limit with a variable, and then take the limit as this variable goes to _____.

Constant

Finite

Infinity

Zero

Question : . By using determinants, we can easily check that the solution of the
given system of linear equation exits and it is unique.
TRUE
FALSE

Question : Let A be an n ×n matrix. The number x is an eigenvalue of A if there exists a non-zero vector
v such that _____.
Av = xv
Ax=xv notshore
Av + xv=0
Av = Ax1

Question : In Jacobi's Method, the rate of convergence is quite _____ compared with other methods.
slow
Fast

Question : .Numerical solution of 2/3 up to four decimal places is _____.

0.667
0.6666
0.6667
0.666671 .

Question : Symbol used for forward differences is
Δ Correct
δ
μ

Question : Muller's method requires --------starting points
1
2
3

While using Relaxation method, which of the following is increment 'dxi' corresponding to the largest Residual for 1st iteration on the system; 2x+3y = 1, 3x +2y = -4 ?

Select correct option:

-2

2

3

4

How many Eigen values will exist corresponding to the function; Exp(ax) = e^ax, when the matrix operator is of differentiation?

Select correct option:

Finite Multiple

Infinite many
Unique

None

Gauss Seidel method is also known as method of Successive displacement

False

True

Secant method falls in the category of

Bracketing method

Open method

Iterative method

Indirect method

For a system of linear equations, the corresponding coefficient matrix has the value of determinant; |A| = -3, then which of the following is true?

The system has unique solution The system has finite multiple solutions

The system has infinite may solutions =>The system has no solution

While solving the system of linear equations; x-y=2, -x+y=3 by jacobi's method, if(0.0) be its first approximate solution, then which of the following is the second approximate solution?

2.3

2.0

0.3

Second solution is not exist.

Q: While solving a system of linear equations by Gauss Jordon Method in which case of the following the partial pivoting is essential?

When the upper triangular entities is zero

When the lower triangular entities is zero

When diagonal entities are zero

When pivots are zero

For the system; 2x+3y = 1, 3x +2y = - 4, if the iterative solution is (0,0) and 'dxi = 2' is the increment in 'y' then which of the following will be taken as next iterative solution?

(2.0) **(0.3)** (0.2) (1.-4)

Q: While solving by Gauss-Seidel method, which of the following is the first Iterative solution for the system;
x-2y =1,
x+4y=4?

=>(1. 0.75) (0.0) **(1.0)** (0.1)

Question : Let A be an n ×n matrix. The number x is an eigenvalue of A if there exists a non-zero vector v such that _____.

Av=xv

Ax=xv

Av+xv=0

Av=Ax1

For the system of equations; x =2, y=3. The inverse of the matrix associated with its coefficients is----------.

singular

non singular

non identity

of order 1*1

In Gauss-Seidel method, each equation of the system is solved for the unknown with -------- coefficient, in terms of remaining unknowns

smallest

largest

any positive

any negative

If there are three equations in two variables, then which of the following is true?

System may have unique solutions

System may have multiple numbers of finite solutions

System may have infinite many solutions

All above possibilities exist depends on the coefficients of variables

The Gaussian elimination method fails if any one of the pivot elements becomes……

Select correct option:

One

Zero

In Jacobi's Method, We assume that the ……elements does not vanish.

Select correct option:

Diagonal

Off-diagonal

Row

Column

In Jacobi's Method, the rate of convergence is quite slow compared with other methods.

TRUE

FALSE

For a function; y=f(x), if y0, y1 and y2 are 2,3 and 5 respectively then which of the following will be 2nd order Backward difference at y2 = 5 ?

-1

-2

1

2

In the context of Jacobi's method for finding Eigen values and Eigen vectors of a real symmetric matrix of order 2*2, if |-5| be its largest off-diagonal then which of the following will be its corresponding off- diagonal values of Orthogonal Matrix?

Cos(theta), -Cos(theta)

Sin(theta), Cos(theta)

Sin(theta), -Sin(theta)

Sin(theta), Cos(theta)

Exact solution of 2/3 is not exists.

TRUE
FALSE

The Jacobi's method is
A method of solving a matrix equation on a matrix that has ____ zeros along its main diagonal.
No
At least one

A 3 x 3 identity matrix have three and _____eigen values.
Same
Different

Eigen values of a symmetric matrix are all _____ .
Real
Complex
Zero
Positive

The Jacobi iteration converges, if A is strictly diagonally dominant.
TRUE
FALSE

If n x n matrices A and B are similar, then they have the same eigenvalues (with the same multiplicities).
TRUE
FALSE

If A is a nxn triangular matrix (upper triangular, lower triangular) or diagonal matrix, the eigenvalues of A are the diagonal entries of A.
TRUE
FALSE

Below are all the finite difference methods EXCEPT _____.
Jacobi's method
Newton's backward difference method
Stirlling formula
Forward difference method

The characteristics polynomial of a 3x 3
Identity matrix is _____, if x is the Eigen values of the given 3 x 3 identity matrix. Where symbol ^ shows power.
(X-1)^3
(x+1)^3
X^3-1
X^3+1

Two matrices with the same characteristic polynomial need not be similar.
TRUE
FALSE

Bisection method is a
Bracketing method
Open method

Regula Falsi means
Method of Correct position
Method of unknown position
Method of false position
Method of known position

Eigenvalues of a symmetric matrix are all _____.
Select correct option:
Real
Zero
Positive
Negative

An eigenvector V is said to be normalized if the coordinate of largest magnitude is equal to zero.
Select correct option:

TRUE
FALSE

Exact solution of 2/3 is not exists.
Select correct option:

TRUE
FALSE

The Gauss-Seidel method is applicable to strictly diagonally dominant or symmetric _____ definite matrices A.
Select correct option:

Positive
Negative

Differences methods find the _____ solution of the system.
Select correct option:

Numerical
Analytical

The Power method can be used only to find the eigenvalue of A that is largest in absolute value—we call this Eigenvalue the dominant eigenvalue of A.
Select correct option:

TRUE
FALSE

The Jacobi's method is a method of solving a matrix equation on a matrix that has no zeros along its _____.
Select correct option:

Main diagonal
Last column
Last row
First row

If A is a nxn triangular matrix (upper triangular, lower triangular) or diagonal matrix , the eigenvalues of A are the diagonal entries of A.
Select correct option:

TRUE
FALSE

A 3 x 3 identity matrix have three and different Eigen values.
Select correct option:

TRUE
FALSE

Newton Raphson method falls in the category of

Bracketing method
Open Method
Iterative Method
Indirect Method

Newton Raphson method is also known as
Tangent Method
Root method
Open Method
Iterative Method

Secant Method uses values for approximation

1
3
2
4

Secant Method is than bisection method for finding root
Slow
Faster

In Newton Raphson method

Root is bracketed
Root is not bracketed

Regula falsi method and bisection method are both

Convergent
Divergent

In bisection method the two points between which the root lies are

Similar to each other
Different
Not defined
Opposite

In which methods we do not need initial approximation to start
Indirect Method
Open Method
Direct Method
Iterative Method

Root may be

Complex
Real
Complex or real

None

In Regula falsi method we choose points that have signs

2 points opposite signs
3 points opposite signs
2 points similar signs
None of the given

In a bounded function values lie between
1 and -1
1 and 2
0 and 1
0 and -2

Newton Raphson method is a method which when it leads to division of number close to zero
Diverges
Converges

Which of the following method is modified form of Newton Raphson Method?
Regula falsi method
Bisection method
Secant method
Jacobi's Method

Which 1 of the following is generalization of Secant method?
Muller's Method
Jacobi's Method
Bisection Method
N-R Method

Secant Method needs starting points
2
3
4
1
Near a simple root Muller's Method converges than the secant method

Faster
Slower

If S is an identity matrix, then

$S^{-1} = S$

$S^{t} = S$

$S^{-1} = S^{t}$

All are true

If we retain r+1 terms in Newton's forward difference formula, we obtain a polynomial of degree ---- agreeing with at

r+2
r+1
R
R-1

P in Newton's forward difference formula is defined as

Octal numbers has the base

10
2
8
16

Newton's divided difference interpolation formula is used when the values of the independent variable are

Equally spaced

Not equally spaced

Constant
None of the above

Given the following data

	0	1	2	4
	1	1	2	5

Value of is

1.5

3
2

1

If is approximated by a polynomial of degree n then the error is given by

Let denotes the closed interval spanned by . Then vanishes ------times in the interval .

N-1
N+2
N
N+1

Differential operator in terms of forward difference operator is given by

---

Finding the first derivative of at =0.4 from the following table:

	0.1	0.2	0.3	0.4
	1.10517	1.22140	1.34986	1.49182

Differential operator in terms of ----------will be used.

Forward difference operator
Backward difference operator
Central difference operator
All of the given choices

For the given table of values

	0.1	0.2	0.3	0.4	0.5	0.6
	0.425	0.475	0.400	0.452	0.525	0.575

, using two-point equation will be calculated as.....

-0.5
0.5

0.75
-0.75

In Simpson’s 1/3 rule, is of the form

- ►
- ►
- ►

While integrating, , width of the interval, is found by the formula-----.

None of the given choices

To apply Simpson’s 1/3 rule, valid number of intervals are.....

7
8
5
3

For the given table of values

	0.1	0.2	0.3	0.4	0.5	0.6
	0.425	0.475	0.400	0.452	0.525	0.575

, using three-point equation will be calculated as ……

17.5
12.5
7.5
-12.5

To apply Simpson’s 1/3 rule, the number of intervals in the following must be

2

3
5
7

To apply Simpson's 3/8 rule, the number of intervals in the following must be

10
11
12
13

If the root of the given equation lies between a and b, then the first approximation to the root of the equation by bisection method is ……

$\frac{(a+b)}{2}$

$\frac{(a-b)}{2}$

$\frac{(b-a)}{2}$

None of the given choices

.....lies in the category of iterative method.

Bisection Method
Regula Falsi Method
Secant Method
All of the given choices

For the equation, the root of the equation lies in the interval.....

(1, 3)
(1, 2)
(0, 1)
(1, 2)

Rate of change of any quantity with respect to another can be modeled by

An ordinary differential equation
A partial differential equation

A polynomial equation

None of the given choices

If

Then the integral of this equation is a curve in

None of the given choices

Xt-plane
Yt-plane
Xy-plane

In solving the differential equation

, By Euler's method is calculated as

1.44
1.11
1.22
1.33

In second order Runge-Kutta method
is given by

$$k_1 = hf(x_n, y_n)$$

None of the given choices

In fourth order Runge-Kutta method, is given by

In fourth order Runge-Kutta method, is given by

None of the given choices

Adam-Moulton P-C method is derived by employing

Newton's backward difference interpolation formula
Newton's forward difference interpolation formula
Newton's divided difference interpolation formula
None of the given choices

The need of numerical integration arises for evaluating the definite integral of a function that has no explicit _____ or whose antiderivative is not easy to obtain

Derivatives
Antiderivative

If then system will have a
Definite solution
Unique solution
Correct solution
No solution

If then
There is a unique solution
There exists a complete solution
There exists no solution
None of the above options

Direct method consists of method
2
3
5
4

We consider Jacobi's method Gauss Seidel Method and relaxation method as
Direct method
Iterative method
Open method

All of the above

In Gauss Elimination method Solution of equation is obtained in
3 stages
2 stages
4 stages
5 stages

Gauss Elimination method fails if any one of the pivot values becomes
Greater
Small
Zero
None of the given

Changing the order of the equation is known as

Pivoting
Interpretation

Full pivoting is than partial pivoting
Easy
More complicated

The following is the variation of Gauss Elimination method

Jacobi's method
Gauss Jordan Elimination method

Courts reduction method is also known as Cholesky Reduction method
True
False

Jacobi's method is also known as method of Simultaneous displacement
True
False
Gauss Seidel method is also known as method of Successive displacement
False
True
In Jacobi's method approximation calculated is used for
Nothing
Calculating the next approximation
Replaced by previous one
All above

In Gauss Seidel method approximation calculated is replaced by previous one
True

False

Relaxation method is derived by
South well
Not defined

Power method is applicable for only
Real metrics
Symmetric
Unsymmetrical
Both symmetric and real

The process of eliminating value of y for intermediate value of x is know as interpolation
True
False

If $A \neq 0$ **then system will have a**

Definite solution

Unique solution

Correct solution

No solution

If $A = 0$ **then**

There is a unique solution

There exists a complete solution

There exists no solution

Then the integral of this equation is a curve in

None of the given choices

Xt-plane

Yt-plane

Xy-plane

solving the differential equation

$$y' = x + y\,; y(0.1) = 1.1$$

$h = 0.1$, **By Euler's method** $y(0.2)$ **is calculated as**

1.44

1.11

1.22

1.33

In Gauss Elimination method Solution of equation is obtained in

3 stages

2 stages

4 stages

5 stages

Which of the following period strategic management was considered to be cure for all problems?
Mid 1950s to mid 1960s
Mid 1960s to mid 1970s
Mid 1970s to mid 1980s Mid
1980s to mid 1990s

Which of the following is not a pitfall an organization should avoid in strategic planning?
Select correct option:
Failing to involve key employees in all phases of planning
Involving all managers rather than delegating planning to a planner
Top managers not actively supporting the strategic planning process
Doing strategic planning only to satisfy accreditation or regulatory requirements

which of the following are the factors that concern the nature and direction of the economy in which a firm operates?
Select correct option:
Technological
Ecological
Social

Economic

Which of the following best describes this statement; "a Systematic and ethical process for gathering and analyzing information about the competition's activities and general business trends to further a business' own goals"?
Select correct option:
External assessment
Industry analysis
Competitive intelligence program
Business ethics

According to Porter, which strategy offers products or services to a small range of customers at the lowest price available on the market?
Select correct option:
Low cost
Best value
Cost focus
Differentiation

Long-term objectives includes all of the following EXCEPT:

Measurable

Reasonable

Varying

Consistent

Which one of the following is NOT is a basic mission of a competitive intelligence program?

To provide a general understanding of an industry

To provide a general understanding of a company's competitors

To identify industry executives who could be hired by the firm

To identify potential moves a competitor might make that would endanger a firm

While preparing an External Factor Evaluation Matrix, a total score of 0.8 indicates that:

Firm is taking advantages of strengths and avoiding threats

Firm is taking no advantage of opportunities and is avoiding threats

Firm is not taking advantages of opportunities and is not avoiding threats

Firm is taking advantage of opportunities and is avoiding the threats

Question # 1 of 10 (Start time: 04:40:08 PM) Total Marks: 1

The determinant of a diagonal matrix is the product of the diagonal elements.

1. **True**
2. False

Question # 2 of 10 (Start time: 04:40:58 PM) Total Marks: 1

Power method is applicable if the eigen vectors corresponding to eigen values are linearly independent.

1. **True**
2. false

A 3 x 3 identity matrix have three and different eigen values.

1. True
2. **False**

If n x n matrices A and B are similar, then they have the different eigenvalues (with the same multiplicities).

1. True
2. **False**

The Jacobi's method is a method of solving a matrix equation on a matrix that has ____ zeros along its main diagonal.

1. **No**
2. At least one

An eigenvector V is said to be normalized if the coordinate of largest magnitude is equal to _____.

1. **Unity**
2. zero

The Gauss-Seidel method is applicable to strictly diagonally dominant or symmetric positive definite matrices A.

1. **True**
2. False

The determinant of a _____ matrix is the product of the diagonal elements.

1. **Diagonal**
2. Upper triangular
3. Lower triangular
4. Scalar

Waisay main nay is ka answer Diagnol keea tha….par yeh charon options theek hain….
You can confirm it from internet…

Jab main yeh MCQ kar raha tha tou tab hi mujhay is par shak ho raha tha….kyun k upper aur lower triangular matrices tou linear algebra mein bhi bahut ziada bataye gaye tou yeh property wahan se hi yaad thi…

Eigenvalues of a symmetric matrix are all _____.

1. **Real**
2. Zero
3. Positive
4. Negative

The Power method can be used only to find the eigen value of A that is largest in absolute value—we call this eigen value the dominant eigen value of A.

1. **True**
2. False

The characteristics polynomial of a 3x 3 identity matrix is _____, if x is the eigen values of the given 3 x 3 identity matrix. where symbol ^ shows power.

1 **(x-1)^3**
2 (x+1)^3
3 x^3-1
4 x^3+1

For differences methods we require the set of values.

1. True

2. False

If n x n matrices A and B are similar, then they have the different eigenvalues (with the same multiplicities).

1. True

2. **False**

If x is an eigen value corresponding to eigen value of V of a matrix A. If a is any constant, then x - a is an eigen value corresponding to eigen vector V is an of the matrix A - a I.

1. True

2. False

a ko agar aap lambda se replace kar dain tou baat clear ho jaye gi….labda ki

jagah a use keea gaya hai tou is liay yeh working thora sa confuse karti hai…

Central difference method seems to be giving a better approximation, however it requires more computations.

1. True
2. **False**

Iterative algorithms can be more rapid than direct methods.

1. **True (main nay true hi keea tha, aap isay** dekh lena)
2. False

Central Difference method is the finite difference method.

1. **True**
2. False

MTH603 MCQs

The determinant of a diagonal matrix is the product of the diagonal elements.

- True
- False

The determinant of a _____ matrix is the product of the diagonal elements.

- Diagonal
- Upper triangular
- Lower triangular
- Scalar

Power method is applicable if the eigen vectors corresponding to eigen values are linearly independent. (Page 6)

- True
- False

Power method is applicable if the Eigen vectors corresponding the Eigen values are linearly

- Independent (Page 6)
- Dependent

Power method is applicable if the Eigen values are real and distinct.

- True
- False

Power method is applicable if the eigen values are _____.

- real and distinct
- real and equal
- positive and distinct
- negative and distinct

A 3 x 3 identity matrix have three and different eigen values.

- True
- False

A 3 x 3 identity matrix have three and _____Eigen values.

- same
- Different

If n x n matrices A and B are similar, then they have the different Eigen values (with the same multiplicities).

- True
- False

If n x n matrices A and B are similar, then they have the _____ eigenvalues (with the same multiplicities).

- same
- different

If n x n matrices A and B are similar, then they have the same eigenvalues (with the same multiplicities).

- TRUE
- FALSE

The Jacobi's method is a method of solving a matrix equation on a matrix that has ____ zeros along its main diagonal. (Bronshtein and Semendyayev 1997, p. 892)

- No
- At least one

The Jacobi's method is a method of solving a matrix equation on a matrix that has no
zeros along its main diagonal.
(Bronshtein and Semendyayev 1997, p. 892).

- True
- False

1.The Jacobi's method is a method of solving a matrix equation on a matrix that has no
zeros along its _____.

- main diagonal
- last column
- last row
- first row

1.An eigenvector V is said to be normalized if the coordinate of largest magnitude is equal to _____.

- Unity
- Zero

An eigenvector V is said to be normalized if the coordinate of largest magnitude is equal to zero.

- TRUE
- FALSE

1.The Gauss-Seidel method is applicable to strictly diagonally dominant or symmetric
positive definite matrices A.

- True
- False

The Gauss-Seidel method is applicable to strictly diagonally dominant or symmetric
definite matrices A.

- Positive
- Negative

Eigenvalues of a symmetric matrix are all _____.

- Real
- Zero
- Positive
- Negative

The Power method can be used only to find the eigenvalue of A that is largest in absolute value—we call this eigenvalue the dominant eigenvalue of A.

- True
- False

The characteristics polynomial of a 3x 3 identity matrix is _____, if x is the eigen values of the given 3 x 3 identity matrix. where symbol ^ shows power.

- (x-1)3
- (x+1)3
- x3-1
- x3+1

1.For differences methods we require the set of values

- True

- False

If x is an eigenvalue corresponding to eigenvalue of V of a matrix A. If a is any constant, then x – a is an eigen value corresponding to eigen vector V is an of the matrix A - a I.

- True
- False

Central difference method seems to be giving a better approximation, however it requires more computations.

- True
- False

1.Iterative algorithms can be more rapid than direct methods.

- True
- False

1.Central Difference method is the finite difference method.

- True
- False

1.The dominant or principal eigenvector of a matrix is an eigenvector corresponding to the
Eigen value of largest magnitude (for real numbers, largest absolute value) of that matrix,

- True
- False

Eigen values of a_____ matrix are all real.

- Symmetric
- Antisymmetric
- Rectangular
- Triangular

Simpson’s rule is a numerical method that approximates the value of a definite integral by using polynomials.

- Quadratic
- Linear
- Cubic
- Quartic

1.In Simpson’s Rule, we use parabolas to approximating each part of the curve. This proves to be very efficient as compared to Trapezoidal rule.

- True
- False

The predictor-corrector method an implicit method. (multi-step methods)

- True
- False

Generally, Adams methods are superior if output at many points is needed.

- True
- False

In Trapezoidal rule, the integral is computed on each of the sub-intervals by using linear interpolating formula, ie. For n=1 and then summing them up to obtain the desired integral.

- True
- False

The Trapezoidal rule is a numerical method that approximates the value of a_____.

- Indefinite integral
- Definite integral
- Improper integral
- Function

The need of numerical integration arises for evaluating the definite integral of a function that has no explicit _____ or whose antiderivative is not easy to obtain.

- Antiderivative
- Derivatives

In Runge – Kutta Method, we do not need to calculate higher order derivatives and find greater accuracy.

- TRUE
- FALSE

1.An indefinite integral may _____ in the sense that the limit defining it may not exist.

- diverge
- converge

1.The Trapezoidal Rule is an improvement over using rectangles because we have much less "missing" from our calculations. We used _____ to model the curve in trapezoidal Rule.

- straight lines
- curves
- parabolas
- constant

An improper integral is the limit of a definite integral as an endpoint of the interval of sintegration approaches either a specified real number or ∞ or -∞ or, in some cases, as both endpoints approach limits.

- TRUE
- FALSE

1.Euler's Method numerically computes the approximate derivative of a function.

- TRUE
- FALSE

1.Euler's Method numerically computes the approximate _____ of a function.

- Antiderivative
- Derivative
- Error
- Value

1.If we wanted to find the value of a definite integral with an infinite limit, we can instead replace the infinite limit with a variable, and then take the limit as this variable goes to _____.

- Constant
- Finite
- Infinity
- Zero

Exact solution of 2/3 is not exists.

- TRUE
- FALSE

The Jacobi iteration converges, if A is strictly diagonally dominant.

- TRUE
- FALSE

1.The Jacobi iteration _____, if A is strictly diagonally dominant.

- converges
- diverges

Below are all the finite difference methods EXCEPT _____.

- jacobi’s method
- newton's backward difference method
- Stirlling formula
- Forward difference method

If A is a nxn triangular matrix (upper triangular, lower triangular) or diagonal matrix, the eigenvalues of A are the diagonal entries of A.

- TRUE
- FALSE

Two matrices with the same characteristic polynomial need not be similar.

- TRUE
- FALSE

Differences methods find the _____ solution of the system.

- numerical
- Analytical

By using determinants, we can easily check that the solution of the given system of linear equation exits and it is unique.

- TRUE
- FALSE

Direct method can more rapid than iterative algorithms

- TRUE
- FALSE

The dominant eigenvector of a matrix is an eigenvector corresponding to the eigenvalue of largest magnitude (for real numbers, smallest absolute value) of that matrix.

- TRUE
- FALSE

The central difference method is finite difference method.

- True
- False

The absolute value of a determinant (|detA|) is the product of the absolute values of the eigenvalues of matrix A

- TRUE

- FALSE

Eigenvectors of a symmetric matrix are orthogonal, but only for distinct eigenvalues.

- TRUE
- FALSE

Let A be an n ×n matrix. The number x is an eigenvalue of A if there exists a non-zero vector v such that _____.

- Av = xv
- Ax = xv
- Av + xv=0
- Av = Ax

In Jacobi's Method, the rate of convergence is quite _____ compared with other

- methods.
- slow
- fast

Numerical solution of 2/3 up to four decimal places is _____.

- 0.667
- 0.6666
- 0.6667
- 0.66667

Euler's method is only useful for a few steps and small step sizes; however Euler's method together with Richardson extrapolation may be used to increase the _____.

- order and accuracy
- divergence

The first langrange polynomial with equally spaced nodes produced the formula for _____.

- Simpson's rule
- Trapezoidal rule
- Newton's method
- Richardson's method

The need of numerical integration arises for evaluating the indefinite integral of a function that has no explicit antiderivative or whose antiderivative is not easy to obtain.

- TRUE
- FALSE

The Euler method is numerically unstable because of _____ convergence of error.

- Slow
- Fast
- Moderate
- No

Adams – Bashforth is a multistep method.

- True
- False

Multistep method does not improve the accuracy of the answer at each step.

- False
- True

1.Generally, Adams methods are superior if output at _____ points is needed.

- Many
- Two

- Single
- At most

Symbol used for forward differences is

-
-
- Δ
- δ
- μ

The relationship between central difference operator and the shift operator is given by

- $\delta = E - E^{-1}$
- $\delta = E + E^{-1}$
- $\delta = E^{1/2} + E^{-1/2}$
- $\delta = E^{1/2} - E^{-1/2}$

Muller's method requires --------starting points

- 1
- 2
- 3
- 4

If we retain r+1 terms in Newton's forward difference formula, we obtain a polynomial
of degree ---- agreeing with y_x at x_0, x_1, ……. X_n.

- r+2
- r+1
- **r**

- r-1

Octal number system has the base ----------

- 2
- 8
- 10
- 16

Newton's divided difference interpolation formula is used when the values of the are

- Equally spaced
- Not equally spaced
- Constant
- None of the above

Rate of change of any quantity with respect to another can be modeled by

- An ordinary differential equation
- A partial differential equation
- A polynomial equation
- None of the given choices

Adam-Moulton P-C method is derived by employing

- Newton's backward difference interpolation formula
- Newton's forward difference interpolation formula
- Newton's divided difference interpolation formula
- None of the given choices

Bisection method is …….. method

- Bracketing Method

- Open
- Random
- none

Newton Raphson method is …….. method

- Bracketing Method
- Open
- Random
- none

Eigenvalue is

- Real
- Vector
- odd
- even

Bisection and false position methods are also known as

- bracketing method
- open method
- random

The Inverse of a matrix can only be found if the matrix is

- Singular
- Non singular
- Scalar
- Diagonal

If f (x) contains trigonometric, exponential or logarithmic functions then this equation is known as

- Transcendental equation
- Algebraic
- Polynomial
- Linear

In interpolation δ is used to represent the
- Forward difference
- Central difference
- Backward difference

The base of the decimal system is _____
- 10
- 0
- 2
- 8
- None of the above

Bisection and false position methods are also known as bracketing method and are always
- Divergent
- Convergent

P in Newton's forward difference formula is defined as
- P=(x-x0)/h
- P=(x+x0/h
- $P=(x+x_n)/h$
- $P=(x-x_n)/h$

Newton's divided difference interpolation formula is used when the values of the are
- Equally spaced

- Not equally spaced
- Constant
- None of the above

Given the following data

X	0	1	2	4
F(x)	1	1	2	5

The value of f(2,4) is

- 1.5
- 3